Pricing of Systematic Liquidity in Government of India Bond Market

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Abstract

System-wide liquidity is an important determinant for the pricing efficiency of any asset market and bond markets are no exception. Research on Indian Bond markets is scarce, and very few studies have examined the impact of idiosyncratic factors on bond returns. This paper, probably the first of its kind, examines the impact of systematic factors like, liquidity, term, remaining maturity and re-issuances on the returns of Government of India bond traded in Indian bond markets. Using the NSE's N-S based ZCYC for the period 2006-2012, this paper finds that the excess bond returns are positively related to systematic liquidity, term risk, and remaining maturity, and negatively related to re-issuances. While all the systematic liquidity factors are statistically significant, the premium for remaining time to maturity and re-issuances does not show any economic significance.

Paper Type: Research Paper

Key Words: Sovereign Bonds, Bond Liquidity, Trading Volume, Age, Volatility, India
JEL Classification: C31, C33, G12

1. Introduction:

System wide liquidity is critical for bond markets. Bond markets are far more illiquid in comparison to equity markets and an average bond trade is of much higher value compared to an equity trade. Lack of overall system wide liquidity can have a serious impact on investor's wealth. In general, increasing overall yields point to diminishing liquidity in the market. This systematic liquidity risk, being un-diversifiable, makes investors demand extra compensation Li et al. (2009). Previous studies on illiquidity of government securities have been examining impact of idiosyncratic factors on bond returns, but not the systematic factors like liquidity. This paper attempts to add to the existing literature on Government of India (GoI) bond markets. Using the market wide liquidity measures proposed by Pastor and Stambaugh (2003) in the Fama-French style factor model (Fama & French, 1993) this paper investigates the impact of system wide liquidity risk, term risk, remaining maturity, and re-issuances on the returns of GoI bonds.

Using National Stock Exchange of India's Nelson-Siegel based ZCYC for the period 2006-2012; this paper finds excess bond returns to be positively related to systematic liquidity, term risk, and remaining maturity, and negatively related to re-issuances. While all the systematic liquidity factors are statistically significant, premiums for remaining time to maturity and re-issuances do not show any economic significance. It also finds that some portion of bond excess return is still not explained by the model presented in this paper.

The rest of the paper is presented in the following sections. Section 2 deliberates on the relevant literature with section 3 presenting the hypotheses. Section 4 explains methodology, variables, model, and data sources followed by section 5 presenting descriptive analysis and the results of the estimated model. Section 6 Concludes.

2. Review of Literature

Muranaga and Ohsawa (1998), defines Liquidity Risk as "the risk of being unable to liquidate a position in a timely manner at a reasonable price". Even Basel Committee (2008) suggests that "Market Liquidity Risk is the risk that a firm cannot easily offset or eliminate a position without significantly affecting the market price because of inadequate market depth or market disruption."

Higher liquidity is identified by higher volume and depth in the market. Brandt and Kavajecz (2004), Easley et al. (2002) also identify order imbalance to affect the liquidity too. Brandt and Kavajecz (2004) attribute daily variation of bond yields to order imbalances observed in GovPix database. Easley et al. (2002), Pastor and Stambaugh (2003) find that information and liquidity risk of the equity returns, derived from trading data, do explain variability of stock return at a significant level. Brandt and Kavajecz (2004) also find evidence of private information in US government securities trading despite having no major macroeconomic announcements. Chordia et al. (2002) use bid-ask spread and depth as main proxy to model liquidity. They find correlation between US stock and bond market return predictability, suggesting that liquidity is indicative of systematic risk.

Amihud and Mendelson (1986b), Vayanos (1998), discusses impact of liquidity risk on equity asset prices. Amihud and Mendelson (1986b) propose theoretical foundation to explain why rational investors will factor their future cost of trading into the trades made today. Investor's factor selling transaction cost while purchasing them. They discount the purchase with respect to anticipated liquidity risk while selling. Investors seek higher returns for higher anticipated selling transaction costs. Warga (1992), Kamara (1994) find that assets with lower liquidity often trade at a discount providing a higher return. Brennan and Subrahmanyam (1996) extend Amihud and Mendelson (1986b) model by including cost for asymmetric information and using Fama and French (1993) to price the liquidity. Amihud and Mendelson (2006) define market liquidity as a trading cost with respect to its fair price. Amihud and Mendelson (2006) define this trading cost as sum of direct trading, price Impact and delay cost.

Information and market microstructure based models hold good for any financial assets, though most of the empirical work has been with stocks. Equity markets have lot more investors trading smaller quantities whereas bond markets have relatively few investors trading larger amount of bonds. Adverse information effect is much stronger in bond trades compared to equity trades. Preference of traders towards a particular target maturity, recently auctioned vs. earlier issued bonds, aging of bonds, choice of coupon amount, tax treatment of trading gains and coupon income are some of the features making bond trading a unique one. These varied trading preferences affect the trading price, hence giving rise to a yield spread over a benchmark. Yield spread is over a particular reference rate and such spread usually accounts for risk effects of liquidity, tax, re-investment, market, asymmetric information, regulatory, and default in case of bonds. These issues make risk for bonds a difficult one to measure as it is entangled by various dimensions of bond trading. Liquidity risk of a bond is often confounded by the above risks making empirical valuation of it a difficult task. Amihud and Mendelson (1991), Fleming (2003), Goldreich et al. (2005), cover aspects of these factors. Periods of financial crisis, like 1998, 2007-08, 2011-12 made bond markets exhibit "run for quality" aspect of liquidity. In addition, government securities issued in local currency also faces inflation risk due to monetizing of the debt. For long tenured sovereign bonds, government's ability to pay the coupon and principal of these bonds depends on macro variables and real economic activities. Chordia et al. (2005) uses such macro variables to determine price of risk.

Any market as a whole poses market liquidity problem. Market liquidity, being varying in nature, also presents a risk being uncertain. This implies that individual asset liquidity is often affected by market wide systematic liquidity. Such phenomenon is termed as commonality in liquidity. Chordia et al. (2000), Hasbrouck and Seppi (2001), Huberman and Halka (2001) provide empirical evidences of this phenomena in US markets. Hasbrouck and Seppi (2001) postulates that markets are neither perfectly liquid nor completely illiquid, hence making liquidity risk partially priced. This leads to presence of such liquidity commonality and pricing of systemic liquidity. Domowitz et al. (2005) identified market liquidity to be uncorrelated with returns by using bid-ask spread as round trip trading cost. They analyze commonality in market liquidity and attribute the absence of correlation due to different economic forces.

Chen et al. (2007) identifies liquidity as a greasing agent for active functioning of the debt market. This active functioning of the market helps the participants in determination of cost of raising debt of various maturities. Typically higher liquidity leads to lower cost of debt. Chen et al. (2007) also argue that liquidity cost prohibits active trading as investors find it hard to continuously hedge their position, hence offer lower price for illiquid securities. Based on LOT measure by Lesmond et al. (1999), they define a new measure using bid-ask spread and zero % return to analyze liquidity of US corporate bonds. Houweling et al. (2005), Chen et al. (2007) find evidence of liquidity pricing in US corporate bond market. Their work supports work of Longstaff (2001) and Longstaff (2004), which indicates that liquidity is appropriately priced in short term and longer tenured US Treasury securities. Mahanti et al. (2008) proposes another measure, "latent liquidity", based on the weighted holding of the bond by various funds. Their claim is that if entities, which are more inclined to trade, hold more of a given bond, they will tend to trade it more. They calculated this measure using the holding and turnover data provided by various US custodians. Another aspect of trading cost is due to counter party search. Searching for a counter party is to achieve better price by minimizing unintended impact cost. Duffie et al. (2005) provides a theoretical model for this purpose.

De Jong and Driessen (2006) used a multifactor model to analyze US bond market illiquidity using bid-ask spread for long tenured US treasury bonds, equity market return, volatility index, and Amihud (2002)'s ILLIQ measure. First step of their model estimates the respective betas and the second step estimates pricing of liquidity risk. They find long term investment grade bonds carrying 0.6% liquidity premium. ILLIQ by Amihud (2002) is one of the widely used measures to assess liquidity risk of an asset in terms of price impact. It is defined as absolute change in asset return per dollar volume of trading. Goyenko et al. (2011) notes liquidity shocks to shift from short to long end of term structure. Beber et al. (2009), Favero et al. (2010) identifies liquidity to be an important factor in case of various European sovereign bond spread over German bunds. Favero et al. (2010) attributes this yield differential to a common default risk factor for Euro bonds only.

Bond liquidity has various aspects with no single universal measure. Researchers have defined and used various proxies and measures to gauge various aspects of liquidity. Since measure of liquidity being uncertain and itself being elusive, it commands premium. Amihud and Mendelson (1986b), Elton and Green (1998), Díaz et al. (2006) have shown extent of premium inbuilt in the trading of illiquid debt securities. Longstaff (2001), Vayanos (2004), Beber et al. (2009), and Goyenko et al. (2011) discuss impact on trading in specific types of bonds due to various economic phases, financial crises leading to phenomenon of "Flight-to-Quality" and "Flight-to-Liquidity". Flight to liquidity becomes the first stop for investor during the periods of economic uncertainty whereas flight to quality is the last stop during periods of economic distress, each with potentially higher liquidity premium. Also during economic distress period, choice of other quality assets becomes quite limited; hence making preference for government securities soar.

Amihud and Mendelson (1986b), Chen (2001) empirically identify that investors do seek higher return when they trade securities involving higher trading cost. Amihud and

Mendelson (1986b) finds empirical evidence of US equity stock return to be a concave function of bid-ask spread. They also show that longer holding horizon attains higher return with a higher bid-ask spread, hence making bid-ask spread a measure of illiquidity. Amihud and Mendelson (1991) show similar decreasing return-spread concave relationship for US Treasuries. Amihud and Mendelson (1991) discusses The Clientele Effect where an investor is OK to hold bonds with wider bid-ask spread for a longer holding period anticipating higher return for illiquid securities. Amihud and Mendelson (1991) also find yield spread of US treasury bonds to have a convex relationship with inverse of remaining time to maturity. Such inverse return and price relationship leads to fact that liquid bonds do have higher price and the extra liquidity attract lower return. Their empirical study finds a correlation of close to 0.97 between the above attributes. Elton and Green (1998) also provide supporting evidence to Amihud and Mendelson (1991) findings. Kamara (1994) provides liquidity spread determinants in case of US Treasury securities. Vayanos (1998) finds clear evidence of investors extending their holding period when transaction costs are high and rising.

In general, due to non-availability of micro-structure level order book data for fixed income securities markets, researchers use other methods to determine variants of bid-ask spread. Chakravarty and Sarkar (1999), Hong and Warga (2000), Kalimipalli and Warga (2002) calculate effective spread as difference of weighed buy and sell prices of same bond on the same day. This method also requires buy/sell identification mechanism of a given order. Schultz (2001) uses difference between trade price and bid price for US corporate bonds and regressing it trade direction dummy variable gets an estimate for round trip transaction cost to be used as a proxy for spread. Lesmond et al. (1999) provides a robust measure, LOT, to capture equity transaction cost, which is strongly correlated with other measures like bid-ask spread. Hong and Warga (2000) discuss effect of trading costs on illiquidity and find it to be endogenous. Vayanos and Weill (2008) discuss a theoretical liquidity model discussing various market factors contributing to illiquidity. Hong and Warga (2000), Schultz (2001), Cohen and Shin (2002), Edwards et al. (2007) identified Time to Maturity, Age, Outstanding Issue Amount as important factors explaining the spread variations. These factors are in addition to Issuer and its credit rating for corporate bonds. According to these studies spread variation is indicative of liquidity.

Chakravarty and Sarkar (1999) studies bid-ask spread behavior in various bond markets of corporate, government and municipal securities. Studies by Fleming (2003), Chris and Gaa

(2004) find bid-ask spread a better liquidity measure, whereas Goldreich et al. (2005) recommends quoted bid-ask spread. These measures are over trade by trade basis and have better explanation in comparison to bond yields. Díaz et al. (2006) uses market share of trading volume to reflect liquidity of a bond. Higher the market share of trading volume higher is the liquidity. They calculate market share of trading volume for a bond as proportion of its volume to the total trading volume. Vayanos and Wang (2007) find illiquidity affecting the price and increased supply making bonds more liquid. Kalimipalli and Warga (2002) finds outstanding amount having no significant effect on liquidity. Amihud and Mendelson (1991), Kamara (1994), Warga (1992), Elton and Green (1998) found positive effect of issue size on bond liquidity leading to higher trading and lower yield in US treasury market. They find this positive effect to a certain extent of issue size being built. Once the trading start declining, higher issue size for the same security leads to higher yield.

Systematic liquidity issues can affect optimal functioning of financial markets. Various liquidity measures are generally correlated across securities trading in a given market. This correlated behavior can be described as systematic liquidity present in that market. Variations in this liquidity can affect overall trades in the market, thus affecting every security. Commonality in liquidity has been studied extensively by Chordia et al. (2000), Huberman and Halka (2001), and Hasbrouck and Seppi (2001) in the context of US equity market. Acharya and Pedersen (2005) identify that adverse system wide liquidity may force an investor to rebalance its portfolio possibly due to unfavorable valuation. In this situation decrease in market wide liquidity may make rebalancing more expensive and hence making investor seek compensation for this system wide liquidity risk (Holmström & Tirole 2001, Amihud 2002, Domowitz et al. 2005, Martınez et al. 2005, Ericsson & Renault 2006, and Sadka 2006).

3. Hypotheses Formulation

GoI bond market has thin trading for various outstanding bonds despite having more than 110 securities of various tenures outstanding. A smaller base of 50 primary dealers participates in the initial auction of government securities. GoI uses frequent reissuances as a primary vehicle to build up the outstanding volume of a dated security. These reissuances lead to increased trading in the market. Considering previous research, empirical evidences, intuition, and the nature of Indian Government Bond market, this paper proposes the following hypotheses.

Hyp 1: *Higher the systematic liquidity-risk, higher the excess return.*
Hyp 2: *Higher the term-risk higher the excess return.*
Hyp 3: *Higher the remaining time to maturity, higher the excess return.*
Hyp 4: *Higher the reissuances, higher market liquidity and lesser the risk.*

4. Research Methodology

As mentioned in the introductory paragraphs, this paper uses market wide liquidity measures, proposed by Pastor and Stambaugh (2003). Their model was originally proposed in the context of equity markets, where outstanding volume of equities is generally taken as constant. This measure is associated with temporary price change caused by order flow. Lower liquidity on a given day tends to trigger stronger next day price reversals based on the (buy or sell) orders for that day. This measure has been used by Li et al. (2009) for the US Treasuries market where treasuries are issued regularly with relatively fixed outstanding volume. GoI bonds, on the other hand, are reopened quite frequently during its lifetime. Since, reopening of the bonds increases the trading quite significantly, this paper uses turnover as the indicator of order flow rather than trading volume. The paper finds that this modified measure does identify the stressed periods of market liquidity accurately.

4.1 Model and Variables

Following the spirit of Fama and French (1993), Gebhardt et al. (2005), and Li et al. (2009), this paper uses cross-section regression of individual bond excess return with various key characteristics of the bonds. In tune with the objective of this paper, the following model in equation '1' is estimated to examine the impact of systematic liquidity, term premium, along with remaining time to maturity and re-issuances on the bond returns.

$$R_t^i - R_t^f = \gamma_{0t} + \gamma_{1t}\beta_{Lt}^j + \gamma_{2t}\beta_{Tt}^j + \gamma_{3t}TTM_t^i + \gamma_{4t}reIssues_t^i + e_t^i \quad \ldots.. 1$$

Each bond 'i' traded in month 't' is considered, and its excess return over the monthly Treasury bill return R_t^f is regressed on systematic liquidity factor (β_{Lt}^j), term factor (β_{Tt}^j), average remaining time to maturity (TTM_t^i), and number of re-issuances in that month

($reIssues_t^i$). The systematic liquidity factor and the term factor are extracted from two other regressions which are explained in detail in sections, 4.1.1, 4.1.2, and 4.1.3. The above model (in equation 1) is regressed for every month and time series of average value of coefficients is estimated with their standard error. Finally Fama-MacBeth error correction (Fama & MacBeth 1973) is applied to determine the impact of volatility of β_{Lt}^j and β_{Tt}^j on excess returns of bonds. Hence $\hat{\gamma_1}$ and $\hat{\gamma_2}$ provide us price of systematic liquidity and term risk respectively.

4.1.1. Term Premium

Apart from liquidity risk, yield for government bonds is substantially influenced by the interest rate differential for the remaining term to maturity. Term premium plays an important role in the bond pricing as it is a proxy for risk of unexpected interest rate changes during the remaining maturity of the bond. Hence term premium is the compensation for holding a bond for a longer period of time in comparison to holding a short maturity bond. Fama and French (1993) found that most of the corporate bond returns can be explained by the term factor along with the default risk factor. Buraschi and Menini (2002) uses yield curve slope (10Yr – 2Yr) for risk determination in repo markets. Dua and Raje (2010) used similar measure to identify determinants of weekly yields of GoI securities, but they used 91 day T-bill yield instead of 2Yr yield.

Term premium for a period "t" is calculated as the difference between 10Yr swap and 2Yr swap rate for that period. Swap rates are used for this purpose because it has comparatively better liquidity. Swap rates for Indian markets were collected from Bloomberg. The following equation in '2' explains determination of term premium.

$$Term_t = Swap_{10Yr,t} - Swap_{2Yr,t} \qquad \ldots\ldots\ldots2$$

Figure-1 shows the movement of swap term premium for the sample period. Swap term premium shows marked increase during the crisis period of late 2008 till 2010. With a declining trend in early 2010, it jumps again during late 2011-12.

<Insert Fig-1 here.>

4.1.2. Systematic Liquidity Measure

Whittaker (2011) uses change in Australian yield curve slope to determine factors for systematic liquidity risk in Australian market. Based on the intuition of next day price reversals for today's order flow and proposed substitution of volume by the turnover, liquidity measure of an individual bond "i" on a given day "j" of a month "t" is computed from the OLS estimate of $\pi i,t$ in the following regression model in equation '3':

$$r^{e}_{i,j+1,t} = \rho_0 + \rho_1 R_{i,j,t} + \pi_{i,t} sign\left(r^{e}_{i,j,t}\right) \frac{vol_{i,j,t}}{osd_{i,j,t}} + \epsilon_{i,d+1,t} \qquad \ldots\ldots.3$$

Where $R_{i,j,t}$ is the bond 'i' return for day 'j' and $r^{e}_{i,j,t} = R_{i,j,t} - R_{b,j,t}$, the excess bond return over equally weighted GoI bond market return. $R_{b,j,t}$ is computed using all sovereign bonds Total Return Index (TRI) from CCIL. *Sign* $(r^{e}_{i,j,t})$ is an indicator of price reversal and is equal to '1' if excess return $r^{e}_{i,j,t}$ is positive else it is equal to '-1', $\pi_{i,t}$ is estimated for bonds having at least 6 trades in a month.

Using the aggregation approach of Chordia et al. (2000) the monthly measure of systematic liquidity is calculated by averaging all the security wide estimates in a month 't' as follows in equation '4'.

$$\hat{\pi}_i = \frac{1}{K} \sum_{k=1}^{K} \pi_{kt} \qquad \ldots\ldots\ldots4$$

Systematic liquidity risk is measured by the unexpected innovations in the system wide liquidity. These innovations affect the trading of all the bonds when system wide liquidity is stressed. The following AR(2) based regression model in equation '5' is estimated for the ϵt as the liquidity innovation.

$$\Delta\hat{\pi}_t = \alpha_0 + \alpha_1 \Delta\hat{\pi}_{t-1} + \alpha_2 \hat{\pi}_{t-1} + \epsilon_t \qquad \ldots\ldots..5$$

System wide liquidity innovations are denoted as Lt in further discussions of this paper. The calculated system wide liquidity series points out periods of distressed liquidity in the last quarter of 2010 and last quarter of 2011. Figure-2 shows spikes in yield spread and yields during those periods. Hence, it is a reliable measure of system wide liquidity.

< Insert Fig-2 here.>

4.1.3. Measuring Systematic Liquidity Beta $\boldsymbol{\beta^{j}_{Lt}}$ & Term Beta $\boldsymbol{\beta^{j}_{Tt}}$

Using the market wide systematic liquidity measure and term premium, liquidity risk $\boldsymbol{\beta^{j}_{Lt}}$ and term risk $\boldsymbol{\beta^{j}_{Tt}}$ is estimated using time series regression following the spirit of Fama and French (1993). Estimated model is specified below in equation '6'

$$R^{i}_{t} - R^{f}_{t} = \alpha^{j}_{0t} + \beta^{j}_{Lt} L_{t} + \beta^{j}_{Tt} Term_{t} + e^{i}_{t} \qquad \ldots\ldots 6$$

Variable L_t is the systematic liquidity and $Term_t$ is the term premium for the month 't'. To estimate the above model five bond trade portfolios are created based on remaining maturity of 0-4Yr, 4-8Yr, 8-12Yr, 12-18Yr and 18-30Yr. For each portfolio, excess monthly return is calculated and regressed using a rolling window of 24 months. The above regression gives a monthly time series of coefficients estimated for each of five portfolios. This time series of coefficients for systematic liquidity and term premium is used to determine price of systematic liquidity risk, and term premium in the main regression model of this paper explained as equation 1 in Section 4.1.

5. Analysis of Results and Discussion

5.1. Descriptive Statistics

Table-1 summarizes various bond characteristics on a quartile basis. In the sample period of 2006-2012, yield has been on an average 7.7033% with 94bp volatility. Empirically yield has a mild negative skew with a median of 7.933%. Spread is 2bp on an average with a volatility of 25bp making yield movement quite volatile. Such volatility is one indication of fair degree of illiquidity in the market. Number of trades shows a right skew indicating market being dominated by fewer securities trading heavily on a regular basis. Similar observation holds true with the trade size indicating liquidity being present for very few bonds trading heavily in the market. More bond trading is there up to age of 5.5 to 6.5 years of age. After that bonds find not much trading. Median remaining maturity is of 8 years, extending up to 75th percentile of 13 years, indicates most of trading being concentrated in this interval making most of GoI debt having short term maturity of around 10 years. Average duration of 5.86 indicates bulk of debt is of around 10-11 year of maturity. High volatility of 3.17 makes maturity on the higher end of the curve. This is because of a lot of old securities are with high coupon and long maturity. Average coupon of 8.35% with an inter-quartile range of 2.12% is indicative of high interest rate environment persisting for extended time. Average outstanding volume is around 28,000cr and a median value of 22,000cr shows more number of bonds with small outstanding volume. Average tradesize of 14.4cr with a median of 5cr indicates illiquidity due to majority of small size trades.

< Insert Table-1 here.>

Table-2 provides correlation matrix between pairs of various characteristics. The upper triangular part of the matrix has the correlation coefficient with p-values in the lower triangular part. There is no significant correlation present between the various characteristics.

< Insert Table-2 here.>

Remaining Maturity is one important factor influencing bond trading. Any government security trades the most in the early cycle of its life where age is low. This translates to higher the remaining maturity, higher the trading. GoI securities have diverse issuance tenures. Average trade value and volatility of various bond characteristics is analyzed conditional to remaining time to maturity and issuance tenure. Table-3 presents the findings. Average yield increases with increasing remaining maturity. Overall average yield, being closest to average yield of “6.5 to 11.5 years” band, indicates market’s trading preference for this remaining time to maturity. 10 year securities are de-facto market choice being major part of this segment. Remaining time to maturity of “More than 11.5 years” has the highest average yield of 8.13%. Overall yield volatility of 94bp is quite high in comparison to yield volatility of 63bp for “6.5 to 11.5 years” and 54bp for “beyond 11.5 years” segment. This supports the stylized fact that higher volatility is inherent with less number of trades of higher trade volume. In fact, the “up to 6.5 years” segment has higher trade volume (25cr) with fewer trades (6.33), higher spread (4bp) and higher trading volume volatility of 38bp. Trades in this segment fall in both the right and left tail of trade volume distribution.

< Insert Table-3 here.>

Overall age of 6.55 years indicates that a typical bond starts experiencing low trading after this age. Bonds with longer remaining maturity trade till the average age of 3.87 years means that these bonds starts experiencing lower trading quite early. In "up to 6.5 years" segment, average age of bond is 9.22 years, indicating bonds of issuance tenure of 15 years or higher appearing for trading. Reissuances of old bonds at later age may be prompting these trades because of low proportion of trading in these bonds compared to lower issuance tenure bonds.

Table-4 presents another perspective of average bond characteristics conditional to Price Impact measure defined by Kalimipalli and Nayak (2012), Kalimipalli et al. (2013), and Helwege et al. (2014). Four univariate portfolios with labels of "Low", "Low Medium", "Medium High" and "High"; are created using quartiles of price impact measure and categorizing each bond trade into one of them based on its price impact. "Low" portfolio has trades of typically large trading volume whereas "High" portfolio consists of small size trades. Portfolio "High" has highest average yield of 7.9% compared to 7.66% for portfolio "Low" but with a lower spread of 0.2bp compared to 6.7bp for portfolio "Low". On an average, portfolio "Low Medium" has higher number of trades (38) with largest trade size (34.74cr) and negative spread of 2.5bp. These trades are mostly of 8-12 years issuance tenure with average coupon of 8.32%. Bonds in this portfolio have average outstanding volume of 31,000cr. High liquidity is evident in these securities having the above characteristics. Rest of the portfolios show marked illiquidity with higher yield and spread. Overall Indian Bond market has ample evidence of being illiquid.

< Insert Table-4 here>

Table-5 provides descriptive statistics for β_L and β_T. β_T is more volatile than β_L indicating term premium taking a lead role in bond returns as average remaining time to maturity being 8.83 years in Indian context. Descriptive statistics of reissuances indicates that reissuances are mostly happening in newly issued bonds and number of newly issued bonds in India is far less than those in the developed markets. As an example, US Treasury auctions sixty new dated securities and reopens various securities forty times in a year on an average. High correlation between β_L and β_T is common as β_L is computed using portfolio construction approach whereas β_T is same for each of these portfolios. Table-6 shows that the correlation of 0.34 between R_f and β_T indicates a high interest rate environment with narrow difference between treasury bills and swap rates in Indian context.

< Insert Table-5 here.>

< Insert Table-6 here.>

5.2. Analysis of Regression Results

Table-7 has the regression results after applying Fama-MacBeth standard error correction.

< Insert Table-7 here.>

β_L coefficient of 0.092 is positive and quite significant indicating that higher systematic liquidity risk does increase the expected return. Unit increase in the system wide liquidity risk will provide for 9.2bp increase in return. Hence hypothesis 1 can’t be rejected even at 0.1% significance level.

β_T coefficient of 0.03 is positive and quite significant indicating that higher term risk does increase the expected return. Unit increase in the higher term risk will provide for 3bp increase in return. Hence hypothesis 2 also can’t be rejected even at 0.1% significance level.

Similarly, TTM and reIssues coefficients have correct signs supporting hypotheses 3 and 4 at more than 0.1% significance level, but their value being very low indicate no premium for these risk factors. Overall 57% of variation in the cross section of the bond returns is explained by the model. Also, α being significant indicates that portion of the excess bond return can still be explained by some extra factor.

6. Conclusion

Systematic liquidity is a significant non-diversifiable risk faced by bond holders. Previous studies examined bond pricing mechanism considering idiosyncratic factors. This paper contributed to the body of knowledge, especially on GoI bond market by empirically investigating whether systematic liquidity risk, along with term risk, remaining time to maturity, and re-issuances are priced in terms of the excess returns of GoI bonds. Using the market wide liquidity measure of Pastor and Stambaugh (2003) and the approach of Li et al. (2009), this paper models the estimation of excess returns of GoI bonds, refining the process with Fama & Mcbeth error (1973) correction. As hypothesized both systematic liquidity risk and term risk depict a positive and statistically significant relationship towards excess returns of bonds. While the other two factors, remaining time to maturity and re-issuances, have also found to be statistically significant, but their value does not suggest any significant economic value for practical use.

Tables

	Min	Q25	Median	Q75	Max	Avg	Std Dev	Skewness	Kurtosis
Yield	2.9970	7.3687	7.9326	8.2943	10.5076	7.7033	0.9402	-1.47	3.13
Spread	-1.4851	-0.0754	0.0163	0.1051	1.4954	0.0173	0.2533	0.26	6.40
Trades	1.0000	1.000	3.0000	11.0000	515.0000	14.8477	33.5797	4.95	33.90
Tradesize	0.0010	0.8000	5.0000	16.6667	425.0000	14.4025	27.5527	5.34	44.20
Age	0.0001	3.3540	5.5296	8.2030	30.5720	6.5539	5.5255	2.06	5.04
Remaining Maturity	0.0060	3.9841	8.0115	13.1509	30.0000	9.6381	7.4861	1.00	0.26
Duration	0.0060	3.3745	5.9610	8.3000	13.1829	5.8644	3.1661	0.02	-1.00
Coupon	4.8300	7.2700	7.9500	9.3900	14.0000	8.3486	1.9183	0.76	-0.40
Outstanding Volume	0.1096	11.5000	22.0000	42.0000	92.0000	27.7625	20.1296	0.69	-0.53

Table 1: Quartile results of various Bond characteristics

	Yield	Spread	Trades	Tradesize	Remaining Maturity	Coupon	Outstanding Volume
Yield		0.25	0.09	-0.34	0.40	-0.14	0.23
Spread	0.00		-0.08	-0.10	-0.09	0.15	-0.04
Trades	0.00	0.00		-0.01	0.17	-0.07	0.18
Tradesize	0.00	0.00	0.40		-0.30	0.07	0.02
Remaining Maturity	0.00	0.00	0.00	0.00		-0.25	0.17
Coupon	0.00	0.00	0.00	0.00	0.00		-0.26
Outstanding Volume	0.00	0.00	0.00	0.09	0.00	0.00	

Table 2: Correlation matrix of various bound characteristics

Upper triangular part of the matrix has correlation coefficients and lower triangular part has p-values.

Attribute Name	Average Values							
	All	Remaining Maturity			Issuance Tenure			
		<6.5 years	6.5-11.5 yrs	> 11.5 yrs	< 8 yrs	8-12 yrs	12-20 yrs	> 20 yrs
Yield	7.7033	7.2929	7.8479	8.1348	7.2511	7.3486	7.8892	8.0465
Spread	0.0173	0.0394	0.0205	-0.0160	-0.0257	0.0075	0.0440	-0.0021
Trades	14.8477	6.3368	21.6223	20.3625	12.9322	17.2937	14.7770	12.7012
Tradesize	14.4025	24.8555	8.5623	5.3757	28.7842	25.4339	8.0007	4.8692
Age	6.5539	9.2249	5.4769	3.8701	3.3049	5.8943	6.5626	9.2353
Outstanding Volume	27.7625	24.9781	30.2073	29.3592	31.8268	30.3241	25.7691	26.2201
	Standard Deviation							
Yield	0.9402	1.1485	0.6309	0.5411	1.2837	1.0477	0.7055	0.7143
Spread	0.2533	0.3099	0.2013	0.2004	0.2795	0.2641	0.2260	0.2713
Trades	33.5797	12.1568	46.1550	37.2740	17.0698	41.9606	34.7953	23.0627
Tradesize	27.5527	37.9604	14.2205	8.2567	34.0164	38.2465	17.0227	10.4560
Age	5.5255	6.6069	4.3653	2.2219	2.4936	3.1691	3.9440	9.4949
Outstanding Volume	20.1296	19.1288	19.9178	21.1749	16.3953	18.3543	20.3031	23.0428

Table 3: Various Bond Characteristics conditional on Remaining Maturity & Issuance Tenure

	Yield	Spread	Trades	Tradesize	Age	Remaining Maturity	Duration	Coupon	Outstanding Volume
Low	7.6607	0.0671	1.2230	13.7804	9.1453	7.4039	4.8603	9.0008	20.8617
Low Medium	7.3993	-0.0248	37.7782	34.7352	5.1359	6.3233	4.2502	8.3192	30.9457
Medium High	7.7963	0.0046	22.6762	11.7609	4.6584	11.6909	6.6896	8.1537	35.5706
High	7.8998	0.0021	5.7068	1.9157	6.3993	12.8980	7.5242	7.7719	25.8769

Table 4: Univariate Portfolio Results based on Price Impact Measure (Kalimipalli & Nayak 2012)

	Min	Median	Avg	Max	Std. Dev.	Skewness	Kurtosis
β_L	-0.0143	0.0008	0.0026	0.0147	0.0076	-0.0290	-0.9364
β_T	-0.0331	0.0256	0.0332	0.1120	0.0349	0.3015	-0.7802
R_J	0.0026	0.0057	0.0051	0.0074	0.0016	-0.2763	-1.4770
Return	0.0025	0.0066	0.0063	0.0084	0.0009	-1.3988	2.0187
TTM	0.0195	6.9139	8.8256	29.9985	7.1756	1.1181	0.5395
Reissues	0.0000	0.0000	0.1642	3.0000	0.4966	3.1622	9.5085

Table 5: Description Statistics of various risk factors

	β_L	***β_T***	***R_J***	***Return***	***TTM***	***Reissues***
β_L		-0.49	-0.07	-0.17	0.02	-0.04
β_T	0.00		0.34	0.45	0.17	0.10
R_J	0.00	0.00		0.68	0.05	0.05
Return	0.00	0.00	0.00		0.40	0,11
TTM	0.17	0.00	0.01	0.00		0.23
Reissues	0.02	0.00	0.01	0.00	0.00	

Table 6: Correlation matrix of various risk factors

α	***β_L***	***β_T***	**TTM**	**Reissues**	**adjR^2**
0.00041*	0.09187**	0.02904***	0.00002***	-0.00003**	0.56511
(2.57)	(3.30)	(3.84)	(11.82)	(-3.45)	

Individual-statistic are in parenthesis, *, **, and *** denote significance at 5%, 1% and 0.1% levels respectively.

Table 7: Fama-MacBeth corrected risk factors coefficients

Figures

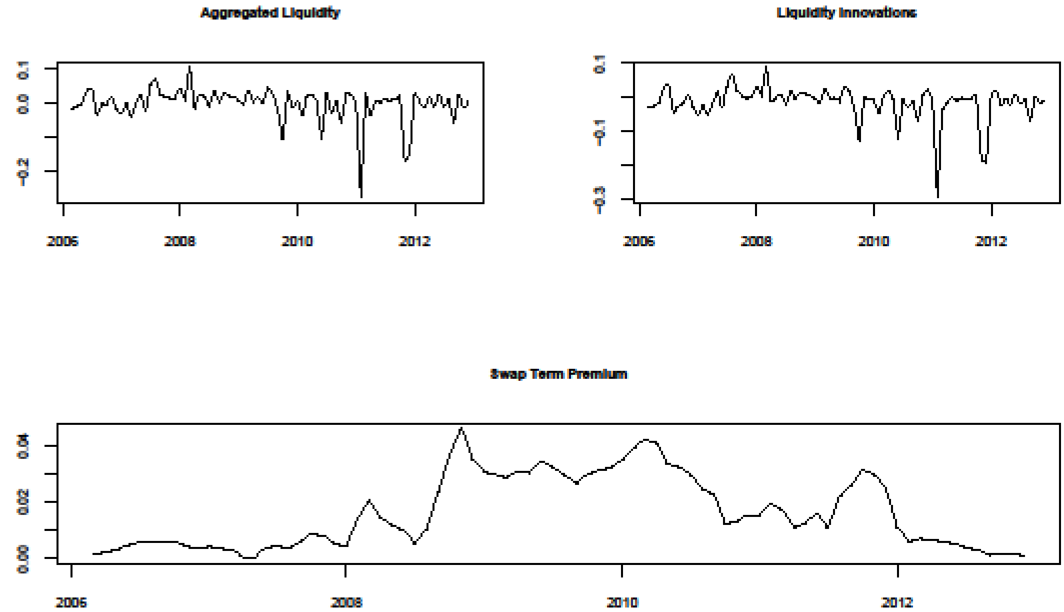

Fig 1: Aggregated Liquidity, Liquidity Innovation and Term Premium

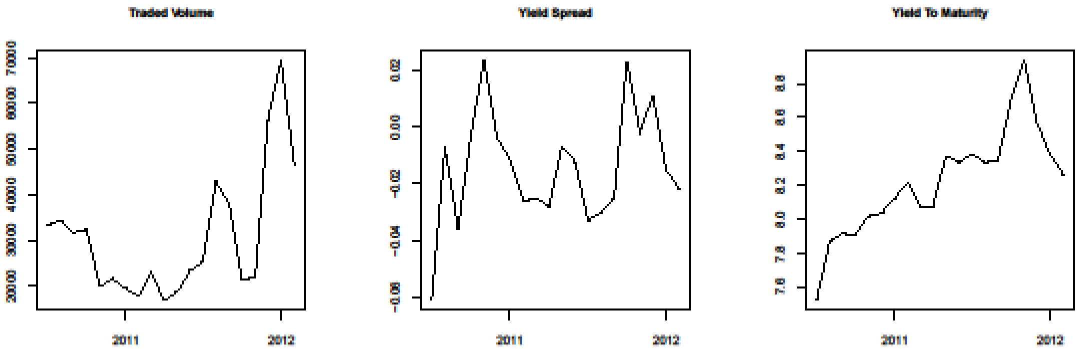

Fig 2: Traded Volume, Yield Spread, and Yield to Maturity around 2011

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